Pet Hotel Management System

Proposal Report

CN230 - Database Systems
Computer Engineering, Thammasat School of Engineering

Covering : Introduction, Stakeholders, Use Cases, Functional Requirements, Database Design Draft, Assumptions & Challenges, and Project Plan

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1. Introduction

1.1 System Overview

Conventional pet boarding facilities predominantly depend on fragmented, paper-based workflows or rudimentary spreadsheet systems to administer their daily operations. This manual approach invariably introduces substantial operational inefficiencies, including concurrent room reservation conflicts, misplaced or outdated vaccination documentation, miscommunication pertaining to individual pet dietary requirements, and inadequate resource utilization tracking. Moreover, pet owners are generally afforded limited real-time visibility into their animals' well-being during the boarding period, a deficiency that contributes to diminished customer satisfaction.

The proposed Pet Hotel Management System is designed to systematically address these shortcomings through a centralized, database-driven architecture. The application aims to digitally transform end-to-end facility operations by establishing well-defined data relationships, enforcing rigorous data integrity, and optimizing front-desk workflows.

1.2 Project Goals and Scope

The primary objective of this project is to design, develop, and implement a robust relational database system capable of supporting the complex operational logic inherent to a modern pet boarding facility.

The scope of the system encompasses the following core functional modules:

- **Customer and Pet Profiling:** Registration and maintenance of owner contact information alongside comprehensive pet medical and vaccination histories.
- **Reservation Management:** Processing of room bookings, real-time availability verification, and add-on service scheduling to prevent temporal conflicts.
- **Billing and Invoicing:** Automated computation of total boarding fees — including base rates and dynamic service charges — accompanied by systematic payment status tracking.
- **Daily Care Monitoring:** Structured logging of daily activities, individualized feeding schedules, and behavioral observations for boarded animals.
- **Inventory Control:** Preliminary tracking and management of essential facility supplies and pet food stock levels.

The project encompasses the conceptual, logical, and physical design of the database; the formulation of optimized SQL queries to support business logic; and the development of a graphical user interface facilitating full Create, Read, Update, and Delete (CRUD) operations.

1.3 Proposed Technology Stack

To ensure the successful development of the Pet Hotel Management System and to maintain strict adherence to academic requirements, the project employs a streamlined, database-centric

technology stack. The selected tools prioritize explicit SQL execution, efficient server-side rendering, and effective team collaboration.

1) Database Tier

- **PostgreSQL** is designated as the primary Relational Database Management System (RDBMS). Its robust, fully ACID-compliant architecture provides native support for complex SQL operations — including multi-table JOINs, subqueries, and advanced aggregations — ensuring efficient management of relational data logic.

2) Backend (Application Logic) Tier

- **Python and Flask** constitute the application server layer. Flask's lightweight micro-framework architecture facilitates direct integration with database adapters such as psycopg2, enabling explicit execution of raw SQL statements without the abstraction introduced by heavy Object-Relational Mappers (ORMs).

3) Frontend (Presentation) Tier

- **HTML with Tailwind CSS via Jinja2** forms the user interface layer. Tailwind CSS enables rapid, utility-first styling, while Jinja2 server-side rendering allows dynamic data retrieved from SQL queries to be injected directly into HTML templates, eliminating the overhead associated with RESTful API development.

4) Development and Management Tools

- **Git and GitHub** are employed for version control and codebase management, with GitHub Projects providing Kanban-based task distribution and milestone tracking.
- **Figma** supports preliminary interface wireframing and user experience planning, while **Canva** is utilized for producing visual assets, architectural diagrams, and presentation materials.

2. System Analysis

2.1 Stakeholders and Their Roles

To accurately capture the system's functional and non-functional requirements, it is crucial to identify the individuals and groups who will interact directly with the database application. The system defines the following primary stakeholders and their respective roles:

- **Pet Owner (Customer):** The primary client utilizing the boarding services. Their system interactions include registering their account, managing pet profiles (specifically verifying medical and vaccination records), making room reservations, selecting add-on services, and reviewing daily pet care reports generated by staff.
- **Receptionist (Front Desk):** The administrative staff responsible for managing the booking pipeline. Their duties involve verifying pet credentials upon check-in, updating room occupancy statuses, processing check-outs, calculating final invoices, and recording financial transactions.
- **Pet Caretaker:** The personnel interacting directly with the boarded animals. They require specialized system access to log daily operational data (such as feeding times, mood, and behavior), execute customized meal plans, and confirm the completion of requested services like grooming.
- **General Staff:** Personnel handling facility maintenance and cleaning. They require access to basic operational schedules and room status updates to maintain hygiene standards across the facility.
- **Hotel Manager:** The administrative personnel overseeing daily floor operations. They require analytical dashboard access to monitor room utilization metrics, track inventory levels (food and supplies), and manage staff user accounts, including role-based access control.
- **Business Owner:** The primary investor or owner of the facility. They require read-only access to high-level strategic data, financial revenue summaries, and system audit trails to ensure operational transparency and guide long-term business growth.

2.2 Use Cases

2.2.1 Module : Booking & Front Desk Management

Use Case 1.1 - Room Reservation

- **Actor:** Pet Owner
- **Pre-condition:** The pet owner possesses an active user account and has registered at least one pet profile within the system.
- **Main Flow:**
    1. The pet owner selects the intended check-in and check-out dates.
    2. The system retrieves and displays all available rooms for the specified period.
    3. The owner selects a room type.
    4. The owner selects any desired add-on services.
    5. The owner confirms the reservation.
- **Post-condition:** The system generates a unique Booking ID and updates the corresponding room status to "Unavailable / Reserved."

Use Case 1.2 - Check-In and Check-Out Processing

- **Actor:** Receptionist
- **Pre-condition:** A confirmed booking record exists within the system.
- **Main Flow:**
    1. The receptionist retrieves the relevant booking record.
    2. The receptionist verifies the pet's information and vaccination history.
    3. The receptionist confirms the check-in or check-out action.
    4. The system updates the room occupancy status accordingly.
- **Post-condition:** Upon check-in, the booking status is updated to "Currently Staying." Upon check-out, the room status is updated to "Available."

Use Case 1.3 - Add-On Service Management

- **Primary Actor:** Pet Owner
- **Secondary Actor:** Receptionist (in the Alternative Flow)
- **Pre-condition:** An active room booking exists in the system, and the associated pet has not yet completed check-out.
- **Main Flow:**
    1. The owner navigates to the "Add-on Services" menu.
    2. The system presents a list of currently available services (e.g., bathing, grooming, massage).
    3. The owner selects a service type, preferred date, and time slot.
    4. The system verifies slot availability and displays the applicable charge.
    5. The owner confirms the service addition.

- **Post-condition:** The system appends the selected service to the booking record, recalculates the total payable amount, and dispatches a confirmation notification to the pet owner.

2.2.2 Module : Billing & Payment Management

Use Case 2.1 — Bill Calculation and Invoice Generation

- **Actor:** Receptionist / Billing System
- **Pre-condition:** The boarded pet is scheduled for check-out.
- **Main Flow:**
    1. The receptionist selects the booking record pending check-out.
    2. The system aggregates all applicable charges, including room fees, add-on service charges, and any emergency veterinary costs where applicable.
    3. The receptionist applies any eligible discounts or promotional codes.
    4. The system computes the net total and renders the result on screen.
    5. The system generates a formal Invoice document for presentation to the customer.
- **Post-condition:** The system records the outstanding payable amount and updates the booking status to "Pending Payment."

Use Case 2.2 - Payment Recording and Receipt Issuance

- **Actor:** Receptionist
- **Pre-condition:** An invoice has been generated, and the customer has confirmed payment at the front desk.
- **Main Flow:**
    1. The receptionist selects the customer's payment method (e.g., cash, QR code bank transfer, credit card).
    2. The receptionist confirms payment receipt within the system.
    3. The system updates the billing status to "Paid."
    4. The system generates and prints an official Receipt for the customer.
- **Post-condition:** The transaction is recorded in the hotel's revenue ledger, and the system authorizes the completion of the check-out process.

2.2.3 Module : Pet care & Daily Reporting

Use Case 3.1 — Daily Care Report Logging

- **Actor:** Pet Caretaker
- **Pre-condition:** The relevant pet is in the "Currently Staying" status.
- **Main Flow:**
    1. The caretaker selects the target pet.
    2. The caretaker enters data pertaining to feeding intake, behavioral observations, and mood assessment.

3. The caretaker uploads accompanying photographs.
4. The caretaker submits and saves the report.

- **Post-condition:** The system persists the daily care report to the database and dispatches a notification to the pet owner.

Use Case 3.2 - Meal Plan Configuration

- **Actor:** Pet Owner
- **Pre-condition:** The pet's profile exists within the system.
- **Main Flow:**
  1. The owner accesses the pet's profile.
  2. The owner creates a new Meal Plan.
  3. The owner specifies food type for each meal period (morning, midday, evening).
  4. The owner defines portion quantities.
  5. The owner saves the meal plan.
- **Post-condition:** The system stores the daily meal plan and associates it with the corresponding pet profile.

2.2.4 Module : Inventory & Business Management

Use Case 4.1 — Food Stock Management

- **Actor:** Manager / General Staff
- **Pre-condition:** A food stock database is operational within the system.
- **Main Flow:**
  1. The user retrieves the complete inventory listing.
  2. The system highlights items with low stock levels or approaching expiration dates.
  3. The user initiates a restocking order or updates current stock quantities.
- **Post-condition:** Food stock quantities are updated, mitigating the risk of shortages or the use of expired food items.

Use Case 4.2 – Supplies Inventory Management

- **Actor:** Manager / General Staff
- **Pre-condition:** A supplies stock database is operational and items have been categorized accordingly.
- **Main Flow:**
  1. The user retrieves the complete supplies inventory.
  2. The user applies category-based filters as required (e.g., cat litter, cleaning agents).
  3. The system highlights items falling below the minimum stock threshold.
  4. The user records a new purchase order, issues supplies for use, or manually updates the remaining stock quantity.

- **Post-condition:** Supplies and consumable stock levels are accurately updated, ensuring continuity of pet care operations.

Use Case 4.3 – Business Overview Dashboard

- **Actor:** Manager
- **Pre-condition:** Sufficient booking and payment data is available within the system.
- **Main Flow:**
    1. The manager selects the desired reporting period.
    2. The system renders categorized revenue summary charts.
    3. Room utilization statistics are displayed.
    4. The manager initiates a report export.
- **Post-condition:** The system generates a comprehensive business summary report file.

2.2.5 Module : Staff & Access Control Management

Use Case 5.1 — Staff Profile and Role Management

- **Actor:** Manager
- **Pre-condition:** The actor is authenticated with Manager-level or Administrator-level privileges.
- **Main Flow:**
    1. The administrator navigates to the Staff Management menu.
    2. The administrator selects "Add New Staff Member."
    3. The administrator enters personal details and contact information.
    4. The administrator assigns a system role (e.g., Receptionist, Veterinarian, Pet Caretaker).
    5. The administrator saves the record.
- **Post-condition:** The system creates a user account linked to the designated role and persists the record to the database.

Use Case 5.2 – System Authentication (Login & Logout)

- **Actor:** All Staff Members
- **Pre-condition:** The staff member possesses a valid user account and password.
- **Main Flow:**
    1. The staff member enters their username and password.
    2. The system validates the credentials and retrieves the associated role.
    3. The user is directed to a role-specific dashboard (e.g., Caretakers are directed to the Meal Plan view; Receptionists to the Booking Management view).
    4. Upon shift completion, the staff member confirms logout.
- **Post-condition:** The system records an access log entry and updates the staff member's status to Online or Offline as appropriate.

Use Case 5.3 – Attendance Time Logging (Clock-In / Clock-Out)

- **Actor:** All Staff Members (including Managers)
- **Pre-condition:** The staff member holds an active user account and is currently authenticated.
- **Main Flow:**
  1. The staff member opens the time recording interface.
  2. The staff member selects "Clock In" at the start of the shift or "Clock Out" at the end of the shift.
  3. The system captures and records the current timestamp.
  4. The system displays a confirmation message upon successful registration.
- **Post-condition:** The attendance record is persisted to the database for subsequent payroll and working-hours computation.

Use Case 5.4 – System Audit Trail Review

- **Actor:** Business Owner
- **Pre-condition:** The system has been actively recording transaction logs (e.g., data deletions, invoice modifications).
- **Main Flow:**
  1. The business owner navigates to the Audit Trail menu.
  2. The business owner applies filters by staff member name, action type, or time range.
  3. The system presents a tabular log detailing which staff member performed which action, on which data record, and at what time.
- **Post-condition:** The business owner obtains the necessary information to verify operational transparency and identify any potential irregularities.

2.2.6 Module : User Account & Profile Management

Use Case 6.1 — Customer Account Registration

- **Actor:** Pet Owner
- **Pre-condition:** The individual does not possess an existing account, and the intended email address has not been previously registered.
- **Main Flow:**
  1. The user accesses the registration page.
  2. The user enters their full name, phone number, email address, and password.
  3. The system dispatches a One-Time Password (OTP) to the provided email or phone number for identity verification.
  4. The user submits the OTP.
  5. The system confirms and activates the account.
- **Post-condition:** The system creates a Customer Account assigned a unique Customer ID and sends a welcome email. The account is immediately available for use.

Use Case 6.2 – Pet Profile Registration and Management

- **Actor:** Pet Owner
- **Pre-condition:** The pet owner is authenticated within the system.
- **Main Flow:**
  1. The owner navigates to "My Pets" and selects "Add New Pet."
  2. The owner enters the pet's details, including name, species, breed, sex, date of birth, weight, and coat color.
  3. The owner uploads a profile photograph.
  4. The owner records the vaccination history (vaccine name, administration date, expiration date).
  5. The owner specifies any pre-existing medical conditions or food allergies.
  6. The owner saves the profile.
- **Post-condition:** The system creates a Pet Profile with a unique Pet ID, associates it with the owner's Customer Account, and persists the record to the database. The profile is immediately available for use in room booking and service selection.

Use Case 6.3 – Notification Management

- **Actor:** Pet Owner (recipient) / System (sender)
- **Pre-condition:** The owner has a pet currently in the "Currently Staying" status, or a notification-triggering event has occurred.
- **Main Flow:**
  1. The system detects a predefined triggering event.
  2. The system selects the configured notification channel (e.g., LINE).
  3. The system dispatches the notification to the pet owner.
  4. The owner receives and reads the notification.

  Automated notification triggers include:
  - Completion of a Daily Care Report by a caretaker, including attached photographs
  - Booking confirmation or cancellation
  - Advance reminder issued one day prior to the scheduled check-in date
  - Payment confirmation, with the Invoice attached
- **Post-condition:** The system logs the notification status as either "Sent" or "Read," and the pet owner may review the full notification history at any time.

2.3 Functional Requirements

FR1 — User & Access Management

- **FR1.1:** The system shall support a customer registration process with identity verification conducted via a One-Time Password (OTP) mechanism, applicable to pet owner accounts.

- **FR1.2:** The system shall provide a role-based authentication interface that directs each authenticated user to a dashboard corresponding to their designated system role.
- **FR1.3:** The system shall permit users holding the Manager role to create, modify, and deactivate staff user accounts.
- **FR1.4:** The system shall record staff attendance by capturing clock-in and clock-out events, each accompanied by a system-generated timestamp.
- **FR1.5:** The system shall maintain a comprehensive Audit Trail that logs all modifications to critical data records — including invoice deletions and booking amendments. Access to this audit log shall be restricted to users holding the Pet Hotel Owner or Manager role.

FR2 — Pet Profile Management

- **FR2.1:** The system shall permit pet owners to create and update pet profiles, encompassing attributes such as name, species, breed, date of birth, and body weight.
- **FR2.2:** The system shall provide a dedicated module for recording vaccination histories, including support for the attachment of veterinary certification documents in PDF format.
- **FR2.3:** The system shall support the documentation of pet-specific medical information, including pre-existing conditions, dietary allergies, and behavioral notes requiring special attention.
- **FR2.4:** The system shall allow pet owners to define and manage individualized daily Meal Plans for each registered pet.

FR3 — Booking & Front Desk Management

- **FR3.1:** The system shall display real-time room availability based on the check-in and check-out dates specified by the user.
    - **FR3.1.1:** The system shall allow customers to select a preferred Room Type (e.g., Standard Fan, Air-Conditioned, VIP Suite) from the list of rooms available within the requested date range.
- **FR3.2:** The system shall automatically calculate the total room charge based on the selected room type and the duration of stay. The computed price shall be locked within the Booking Record at the time of reservation confirmation and shall remain unchanged regardless of any subsequent modifications to the room's base rate.
- **FR3.3:** The system shall support the addition of Add-on Services (e.g., bathing, grooming) both at the time of initial booking and during an active stay.
- **FR3.4:** The system shall provide Check-In and Check-Out functions that update both the room occupancy status and the corresponding booking status upon execution.

- **FR3.5:** The system shall allow users holding the Administrator role to add and modify room records — including room number, room size, and base rate — as well as update room operational status (e.g., Available, Under Cleaning, Under Maintenance).
- **FR3.6:** The system shall allow pet owners to cancel a booking prior to the scheduled check-in date. Upon cancellation, the system shall automatically update the booking status to "Cancelled" and revert the corresponding room status to "Available."
    - **FR3.6.1:** The system shall record the date, time, and identity of the user who initiated the cancellation, distinguishing between a pet owner and a receptionist.

FR4 — Pet Care & Monitoring

- **FR4.1:** The system shall provide a Daily Report entry function, enabling caretakers to log records pertaining to feeding intake, bowel activity, and behavioral mood for each boarded pet.
- **FR4.2:** The system shall automatically dispatch a notification to the pet owner upon the submission of each daily care report update.
- **FR4.3:** The system shall send an automated reminder notification to the pet owner one day in advance of the scheduled check-in date.

FR5 — Billing & Payment Management

- **FR5.1:** The system shall aggregate all applicable charges — including room fees and add-on service costs — to generate a consolidated Invoice document.
- **FR5.2:** The system shall support the recording of payments across multiple accepted methods, including cash, QR PromptPay, and credit card transactions.
- **FR5.3:** Upon a payment status being updated to "Paid," the system shall generate an official Receipt in a printable format or as an exportable PDF document.

FR6 — Inventory & Analytics Management

- **FR6.1:** The system shall automatically deduct food and supply stock quantities based on recorded consumption entries.
    - **FR6.1.1:** The system shall trigger an automated alert notification when the remaining stock level of any item falls below 20% of its total registered quantity.
- **FR6.3:** The system shall present a management Dashboard summarizing revenue figures, room occupancy statistics, and add-on service sales performance, accessible to users with managerial or executive privileges.

FR7 — Notification Management

- **FR7.1:** The system shall automatically dispatch a notification to the relevant pet owner whenever the status of a booking record undergoes a change, including events such as booking confirmation and booking cancellation.
- **FR7.2:** The system shall immediately notify the assigned receptionist upon the creation of a new booking record within the system.

3. Draft Database Design

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Asd

4. Assumptions and Anticipated Challenges

4.1 Design Assumptions

The following assumptions have been established to define the operational boundaries of the system and to guide all subsequent design and development decisions.

4.1.1 — Single Pet per Booking Record Each booking record is associated with one pet only. In cases where a single owner wishes to board multiple pets simultaneously, the system shall generate separate and independent booking records for each individual pet.

4.1.2 — Daily-Level Room Availability Management Room availability is managed at the granularity of a full calendar day. The system does not support hourly or time-slot-based booking. Check-in and check-out are processed on a per-day basis.

4.1.3 — Internal Staff Access Only In the initial release phase, system access is restricted exclusively to internal staff, including receptionists, pet caretakers, general staff, managers, and the business owner. Customer-facing login and self-service functionalities are not included within the current project scope and are designated for a future development phase.

4.1.4 — Free-Text Medical and Vaccination Records Vaccination histories and medical information are stored as free-text notes entered manually by staff. The system does not implement a structured medical record schema in this phase. Attachment of veterinary certification documents in PDF format is supported as supplementary evidence.

4.1.5 — External Payment Processing The system does not integrate with any payment gateway or financial processing service. All monetary transactions are handled through external means (e.g., cash, QR PromptPay, credit card terminal). The system records only the payment status (Paid / Unpaid) and the payment method for billing and reporting purposes.

4.1.6 — Fixed Room Rate at Time of Booking The total room charge is calculated and locked at the time of booking confirmation based on the room's base rate at that moment. Subsequent modifications to the room's base rate shall not retroactively affect any previously confirmed booking records.

4.2 Expected Challenges and Limitations

4.2.1 — Concurrent Booking Conflicts and Race Conditions When two or more staff members attempt to reserve the same room for an overlapping date range simultaneously, the system is susceptible to double-booking errors. This challenge requires the implementation of transaction-level locking mechanisms within the PostgreSQL database — specifically, the use of SELECT FOR UPDATE or serializable transaction isolation — to ensure that room availability checks and reservation insertions are executed as atomic operations.

4.2.2 — Data Completeness and Input Quality Pet medical histories, vaccination records, feeding preferences, and behavioral notes are entered manually by staff. Inconsistent or incomplete data entry represents a persistent risk to data quality. While the system enforces required fields and basic validation rules, these measures cannot fully guarantee the accuracy or completeness of free-text inputs. Staff training and standardized data entry protocols will be essential to mitigating this limitation.

4.2.3 — Scope Creep and Feature Boundary Management Several features were discussed informally during the requirements gathering phase, including customer-facing online booking, automated SMS and email reminder notifications, LINE integration, and a customer loyalty rewards program. These features carry significant implementation complexity and are explicitly excluded from the current release scope. Without clearly defined boundaries, incremental feature additions risk destabilizing the project timeline and compromising the quality of core deliverables.

4.2.4 — Notification Delivery Reliability The system relies on external notification channels — particularly LINE — for dispatching daily care report updates, booking confirmations, and reminder alerts to pet owners. Delivery reliability is subject to third-party service availability and network conditions, which fall outside the system's direct control. Failures in notification delivery may affect customer satisfaction and trust in system responsiveness.

5. Project Plan

5.1 Timeline & Work schedule

Phase	Activities	Duration
Phase 1 – Requirements & Analysis	Finalize stakeholder requirements, validate all use cases across the six modules, confirm functional requirements (FR1–FR7), and obtain supervisor approval on project scope	Week 1 – 2
Phase 2 — Database Design	Develop the complete E/R diagram, normalize the relational schema to Third Normal Form (3NF), define all primary and foreign key constraints, and design optimized SQL query structures for core business logic	Week 3 – 4
Phase 3 — UI/UX Design	Produce high-fidelity wireframes in Figma for all role-specific dashboards (Pet Owner, Receptionist, Caretaker, Manager, Business Owner), and design visual assets using Canva	Week 4 – 5
Phase 4 — System Development	Implement the PostgreSQL database schema, develop the Flask backend with raw SQL via psycopg2, and build all HTML/Tailwind CSS frontend templates rendered through Jinja2	Week 5 – 9
Phase 5 — Testing & Quality Assurance	Conduct unit testing on all SQL queries and Flask routes, perform integration testing across all six functional modules, and execute user acceptance testing against the defined success criteria	Week 10 –11
Phase 6 — Deployment & Documentation	Deploy the completed application, conduct team review, finalize all technical documentation, and prepare the project submission package	Week 12

5.2 Team Member Responsibilities

The project team consists of four members, each assigned a primary area of ownership. Responsibilities are distributed to align with the system's architectural layers while ensuring cross-functional collaboration throughout all development phases.

Member	StudentID	Primary Role
Chotiwit Daugsatan	6710615060	Frontend Developer
Purich Ampawa	6710615185	Project Lead & Backend Developer
Athiphat Sunsit	6710615292	Frontend Develop & QA Engineer
Theepop Rattanasubsiri	6710685014	Database Designer

All team members are jointly responsible for code review, GitHub Projects task tracking, and the preparation of submission deliverables.

6. Conclusion

6.1 Summary of Next Development Steps

This proposal has established the foundational analytical and architectural groundwork for the Pet Hotel Management System. The system has been scoped across six functional modules — Booking & Front Desk Management, Billing & Payment Management, Pet Care & Daily Reporting, Inventory & Business Management, Staff & Access Control Management, and User Account & Profile Management — with thirty-one use cases and twenty-six functional requirements formally documented across FR1 through FR7.

The immediate next steps are as follows. First, the relational schema introduced in this document must be refined to full Third Normal Form, with all entity attributes, domain constraints, and referential integrity rules explicitly defined. Second, the Figma wireframing process must be initiated in parallel with schema finalization, ensuring that database design decisions are reflected accurately in the user interface layout for each of the five distinct user roles. Third, the PostgreSQL database instance must be provisioned and the initial Data Definition Language (DDL) scripts must be authored and version-controlled within the team's GitHub repository.

Following the completion of the design phase, the development effort will proceed in module priority order: the authentication and user management module (FR1) will be implemented first as it underpins all role-based access control throughout the system; the booking and front desk module (FR3) will follow as the operationally critical core of the application; and the remaining modules — billing (FR5), pet care (FR4), inventory (FR6), and notification management (FR7) — will be developed and integrated in subsequent iterations.

The system will be subject to structured testing against the success criteria defined in Section 5 of this proposal, with particular attention to the correctness of SQL query outputs, the enforcement of booking conflict prevention logic, and the accuracy of automated invoice computation. Upon successful completion of all testing phases, the application will be packaged for final academic submission in accordance with the requirements of CN230 — Database Systems, Thammasat School of Engineering.